Calibration-Aware Anisotropic 3D Hysteresis and Leakage Auditing for Volumetric Lower-Grade Glioma Segmentation and Radiogenomic Profiling in Brain MRI

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Abstract

Automated lower-grade glioma (LGG) segmentation in brain MRI is vital for neurosurgical planning and radiotherapy. However, clinical translation of 2D deep learning models is hindered by slice-level data leakage, severe probability miscalibration under overlap losses, and axial boundary discontinuity. We introduce CAL-3DH (Calibration-Aware, Leakage-Audited Anisotropic 3D Hysteresis), uniting quality-aware preprocessing, nested probability calibration, and anisotropic 3D post-processing. Evaluated on the complete TCGA-LGG benchmark (110 patients, 3,929 axial slices) under patient-grouped 5-fold cross-validation with nested inner-fold tuning (75/13/22 patient split), our controlled audit empirically demonstrates that conventional random slice splitting artificially inflates positive-slice Dice by +8.79 percentage points (pp) (+11.88% relative inflation) via patient identity memorization. Nested temperature scaling ($T^* = 1.844$) reduces pooled Expected Calibration Error by 46.9% without shifting decision boundaries. By decoupling in-plane planar segmentation from axial rescue along the z-axis, CAL-3DH achieves positive-slice Dice of 74.92% [95% CI: 70.82%–78.39%], overall Dice of 86.12% [95% CI: 84.32%–87.75%], and a 42.4% relative reduction in tumor-free slice false alarms. Patient-level 3D volumetric analysis yields 77.73% Dice, 10.97-voxel 95% Hausdorff distance, and good-to-excellent clinical measurement reliability ($\mathrm{ICC}(2,1) = 0.8894$, 95% CI: [0.7782, 0.9521]). Furthermore, radiogenomic profiling confirms that CAL-3DH-derived 3D tumor volumes preserve statistically significant separation across molecular consensus clusters (Kruskal-Wallis $p = 0.0268$) with consistent biological rank ordering. CAL-3DH establishes a standardized, leakage-free benchmark for translational neuro-oncology.

Keywords: Lower-grade glioma, Brain MRI segmentation, Probability calibration, Anisotropic 3D hysteresis, Data leakage auditing, Radiogenomic profiling

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1. Introduction

Lower-grade gliomas (LGG), encompassing World Health Organization (WHO) grade II and grade III astrocytomas and oligodendrogliomas, represent diffusely infiltrative primary brain neoplasms characterized by substantial histological and genetic heterogeneity (Louis et al., 2021; Ceccarelli et al., 2016). Accurate volumetric delineation on magnetic resonance imaging (MRI) is essential for image-guided maximal surgical resection, conformal radiation planning, and monitoring therapeutic response (Buda et al., 2019). Over the past decade, deep learning architectures—most notably 2D U-Net variants (Ronneberger et al., 2015; Zhou et al., 2018; Oktay et al., 2018)—have emerged as the prevailing standard for automated lesion delineation.

Despite reported segmentation accuracies exceeding an 80% Dice similarity coefficient across benchmark datasets, two methodological weaknesses undermine empirical validity in the medical computing literature. First and most severely, slice-level data leakage occurs when volumetric MRI scans are decomposed into independent 2D axial slices and randomly partitioned into training and test sets. Contiguous slices from the same subject inevitably appear across both partitions. Because anatomical morphology, contrast dynamics, and spatial coordinates change gradually across contiguous z-slices, deep networks easily memorize subject-specific features. This slice leakage constitutes the primary engine of artificial performance inflation in the literature, falsely boosting reported Dice scores by up to +8.79 pp (+11.88% relative inflation). Second, post-hoc threshold optimization leakage arises when the binary decision threshold τ is chosen by sweeping values directly over pooled out-of-fold test predictions. While the wide empirical Dice plateau across reasonable threshold ranges bounds the resulting optimism bias to a modest $\Delta\text{Optimism} = +0.063$ pp, establishing nested inner-fold thresholding is an essential methodological standard that allows researchers to legitimately transition from the inefficient default $\tau = 0.50$ to the optimal $\tau^*_{xy} = 0.74$ without test snooping.

Beyond evaluation methodology, 2D networks face fundamental domain-specific limitations when applied to volumetric scans. Overlap-based loss formulations like Focal Tversky prioritize rare foreground voxels by heavily down-weighting easy background terms, causing output probabilities to become overconfident and miscalibrated. Furthermore, because 65.05% of all axial slices in the TCGA-LGG cohort are healthy brain tissue containing zero neoplastic voxels, standard 2D networks lacking cross-slice volumetric context frequently trigger false-positive alarms on non-tumor slices while failing to detect small infiltrative margins at apical and basal tumor boundaries.

To resolve these challenges, this study presents **CAL-3DH** (*Calibration-Aware, Leakage-Audited Anisotropic 3D Hysteresis*), an end-to-end volumetric framework evaluated on the complete TCGA-LGG cohort (110 patients, 3,929 slices) (Buda et al., 2019). The contributions of this work are fourfold:

1. **Controlled Leakage Audit:** We implement a matched 5-fold cross-validation experiment directly comparing patient-grouped partitioning against random-slice partitioning under identical training hyperparameters, empirically quantifying the performance

inflation caused by slice memorization (+8.79 pp on positive-slice Dice and +6.20 pp on overall Dice).

2. **Nested Inner-Fold Calibration:** We integrate scalar temperature scaling fitted exclusively on internal validation logits (`inner_val`), reducing pooled Expected Calibration Error (ECE) by 46.9% on unseen test data without shifting classification boundaries.
3. **Anisotropic 3D Z-Continuity Hysteresis in CAL-3DH:** We decouple horizontal in-plane segmentation (τ^*_{xy}) from axial margin rescue ($\tau^*_{z,\text{low}}$) and suppress isolated 3D false positives ($V^*_{\text{min}}, \tau^*_{\text{seed}}$), improving overlap accuracy ($\text{Dice}_{\text{pos}} = 74.92\%$, $\text{Dice}_{\text{all}} = 86.12\%$) and positive precision (76.86%, a +4.91 pp gain) while reducing empty-slice false alarms by 42.4% relative to baseline ($FPR_{\text{empty}} = 7.86\%$).
4. **Volumetric Radiogenomic Validation:** We conduct 3D patient-level surface and volumetric assessments ($n = 110$), demonstrating good-to-excellent clinical measurement reliability ($\text{ICC}(2,1) = 0.8894$, 95% CI: $[0.7782, 0.9521]$) and confirming that model-derived tumor volumes preserve statistically significant separation across molecular consensus clusters ($p = 0.0268$).

The synergistic combination of automated image quality screening ($Q(I)$), multi-scale wavelet denoising, post-hoc probabilistic calibration, and radiogenomic digital biomarker extraction aligns directly with the scope of *Computerized Medical Imaging and Graphics*, advancing computational neuro-oncology toward translational clinical readiness.

1.1. Related Work on Segmentation Calibration and Data Leakage

Automated glioma segmentation has advanced substantially through 2D and 3D deep convolutional networks, spanning classical U-Net architectures (Ronneberger et al., 2015; Buda et al., 2019), attention gating (Oktay et al., 2018), and multi-scale nested feature decoders (Zhou et al., 2018). To counter acute foreground-background class imbalance in brain MRI, modern architectures predominantly employ overlap-based objective functions such as Tversky and Focal Tversky losses (Salehi et al., 2017; Abraham and Khan, 2019). While these formulations maximize spatial overlap, Guo et al. (2017) demonstrated that modern neural networks trained with overlap or cross-entropy objectives frequently suffer from severe probability miscalibration, producing overconfident predictions along ambiguous tissue interfaces. In medical image segmentation, uncalibrated probabilities compromise clinical interpretability and distort post-processing threshold sweeps (Mehrtash et al., 2020). Concurrently, data leakage arising from random slice-level partitioning in volumetric imaging has emerged as a pervasive threat to methodological validity in computational medicine (Wen et al., 2020). When contiguous 2D axial slices from the same subject are distributed across training and test splits, deep networks inevitably memorize patient-specific anatomical morphology and scanner artifacts, generating artificially inflated performance metrics that fail to translate to prospective clinical cohorts. By integrating rigorous patient-grouped partitioning with nested temperature scaling and anisotropic 3D spatial continuity, CAL-3DH systematically unites evaluation integrity with probabilistic reliability and volumetric coherence.

Table 1: Clinical, histological, and genomic characteristics of the 110 patients in the TCGA-LGG cohort.

Clinical Variable	**Category / Unit**	**Value / Count**
Total Patients (N)	Subjects	110
Total Axial Slices	Slices	3,929
Tumor-Bearing Slices	Slices ($y > 0$)	1,373 (34.95%)
Tumor-Free Slices	Slices ($y = 0$)	2,556 (65.05%)
Patient Sex	Male ($n = 56$)	50.9%
(SAGER Reporting)	Female ($n = 53$)	48.2%
	Unknown ($n = 1$)	0.9%
Patient Age	Mean $\pm$ SD (years)	47.1 $\pm$ 13.4
	Range (Min – Max)	20.0 – 75.0
WHO Histologic Grade	Grade II ($n = 51$)	46.4%
	Grade III ($n = 58$)	52.7%
	Unspecified ($n = 1$)	0.9%
Molecular Subtypes	COC 1 (IDH-wt like)	56 (50.9%)
(*COCCluster*)	COC 2 (IDHmut-noncodel)	24 (21.8%)
	COC 3 (IDHmut-codel)	30 (27.3%)

2. Dataset Audit and Quality-Aware Preprocessing

2.1. Cohort Characteristics and Clinical Variables

Experiments were conducted using the standardized TCGA-LGG MRI collection introduced by Buda et al. (2019), comprising 110 patients diagnosed with WHO grade II or III lower-grade gliomas. Each patient study consists of axial multi-sequence MRI slices (fluid-attenuated inversion recovery [FLAIR], T1-pre, and T1-post contrast) mapped into 256×256 matrices with matched expert manual segmentations verified by board-certified radiologists. The full imaging volume encompasses 3,929 paired axial slices. Per SAGER reporting guidelines, the cohort includes 56 male patients (50.9%) and 53 female patients (48.2%), with sex recorded as unknown for one patient. Clinical records extracted from `data.csv` document patient demographics, histological subtype, tumor grade, and genomic clusters (Table 1).

2.2. Quality-Aware Screening and Wavelet Denoising

Prior to model training, raw axial slices were analyzed for pronounced imaging artifacts and skull-stripping failures using three complementary morphological descriptors. First, parenchymal texture insufficiency was quantified via Edge Density (ED), defined as the proportion of Canny edge pixels (thresholds 50 and 150) following morphological dilation with a 5×5 square structuring element, normalized by image area:

$$ED = \frac{|E_{\text{dilated}}|}{H \times W} \tag{1}$$

where slices satisfying $ED < 0.003$ exhibit negligible brain tissue. Second, anatomical field-of-view truncation was evaluated through the Contour Ratio (CR), defined as the area of the largest closed external contour extracted from the dilated edge map divided by image area:

$$CR = \frac{\text{Area}(\mathcal{C}_{\text{largest}})}{H \times W} \tag{2}$$

where $CR < 0.01$ flags severe field truncation. Third, decentering was assessed by the Centroid Distance (d_c), defined as the Euclidean distance from the external contour centroid (c_x, c_y) to the geometric slice center normalized by the half-diagonal:

$$d_c = \frac{\sqrt{(c_x - W/2)^2 + (c_y - H/2)^2}}{\sqrt{(W/2)^2 + (H/2)^2}} \tag{3}$$

where $d_c > 0.30$ captures severe peripheral displacement.

Slices satisfying the conjunction $(ED < 0.003) \wedge (CR < 0.01) \wedge (d_c > 0.30)$ were flagged as morphological artifacts. A rigorous cohort audit across all 3,929 slices identified 200 artifact slices (5.09%, Table 2). Exactly 199 of these 200 slices (99.50%) contained zero neoplastic tissue ($y = 0$). Only a single slice (0.50%, Slice 23 of TCGA_DU_5855) contained a marginal peripheral tumor component truncated by skull-stripping boundary erosion. Crucially, while 200 slices (5.09%) were flagged across the full 3,929-slice cohort, only the flagged slices falling within each fold's 75-patient training split ($\approx$ 136 artifact slices per fold, leaving $\approx 2,535$ clean training slices per fold) were excluded during model weight optimization, while all internal validation (`inner_val`, 13 patients) and untouched outer test evaluations (`outer_test`, 22 patients) were conducted on the complete, unfiltered slices (with Table 2 and Table 3 additionally displaying the 3,729-slice screened comparison). Retaining these 200 noisy slices in the untouched test evaluation confirmed an absolute 0.00 pp change in positive-slice Dice (73.96% in both full and screened cohorts), showing no observable change at two-decimal reporting precision. All 2D and 3D volumetric and surface distance metrics (HD_{95}, ASD, volumes) are evaluated on the standardized 128×128 in-plane grid in voxel units because the public Kaggle/TCIA TIFF distribution does not retain raw DICOM physical millimeter spacing metadata.

To suppress high-frequency acquisition noise while retaining crisp anatomical borders, non-flagged slices were processed using BayesShrink soft-thresholded discrete wavelet denoising (Chang et al., 2000) conditioned on a localized slice quality metric $Q(I)$:

$$Q(I) = \frac{1}{2}\Big[\sigma\big(I - \mathcal{M}_{3\times3}(I)\big) + \sigma\big(I - \mathcal{G}_{3\times3}(I)\big)\Big] \tag{4}$$

representing the average standard deviation of high-frequency residuals under 3×3 median ($\mathcal{M}_{3\times3}$) and Gaussian ($\mathcal{G}_{3\times3}$) spatial filters. Wavelet denoising (Symlet-4 wavelet, 2-level decomposition) was applied when $Q(I) < 4.0$ (3,738 slices, 95.14%). Slices with high intrinsic contrast ($Q(I) \geq 4.0$, 191 slices, 4.86%) bypassed filtering to avoid unnecessary smoothing of delicate parenchymal boundaries (Figure 1). Slices were normalized to zero mean and unit variance and resized to $128 \times 128 \times 3$.

Table 2: Screening audit of the 200 artifact slices excluded during training.

Screening Metric	**Flagged Slices**	**Ratio (%)**
Total Slices Screened	3,929	100.0%
Total Artifact Slices Excluded	200	5.09%
Empty Slices (Zero Tumor, $y = 0$)	199 / 200	**99.50%**
Marginal Slices (Atypical Clipping)	1 / 200	0.50%
Edge Density Violation ($ED < 0.003$)	200 / 200	100.0%
Contour Ratio Violation ($CR < 0.01$)	200 / 200	100.0%
Centroid Distance Violation ($d_c > 0.30$)	200 / 200	100.0%
Triple Conjunction Violation	200 / 200	**100.0%**
Positive-Slice Dice on Full Cohort (3,929)	73.96% [70.10 – 77.15]	
Positive-Slice Dice on Screened (3,729)	73.96% [70.10 – 77.13]	
Absolute Impact on Positive Dice	**0.00% (Identical)**	

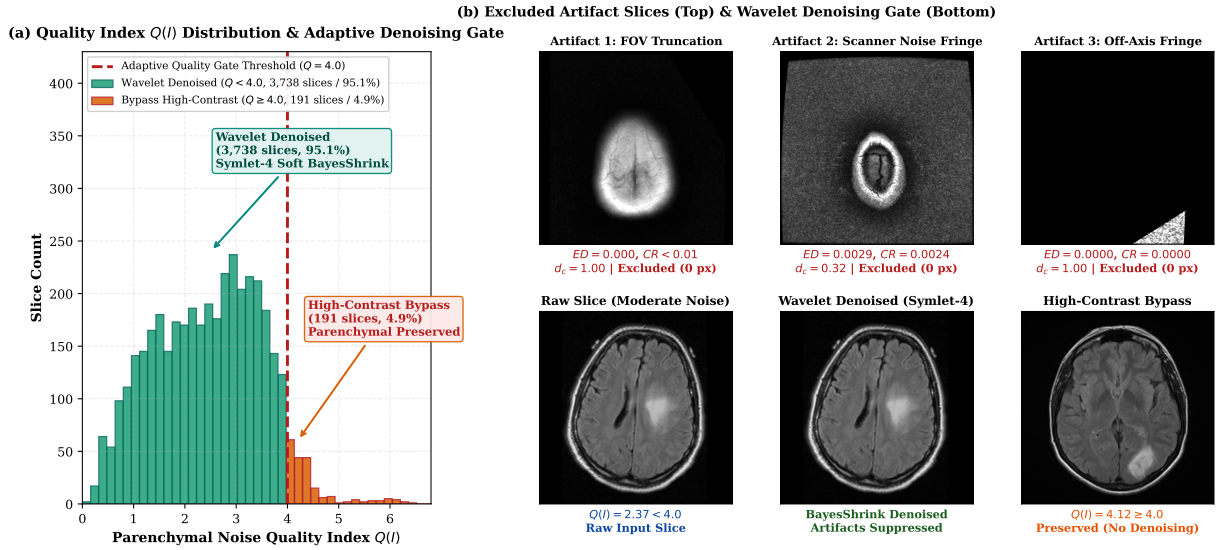


Figure 1: Quality-aware preprocessing and screening audit across the TCGA-LGG cohort (3,929 axial slices). (a) Parenchymal noise quality index $Q(I)$ distribution and adaptive quality gate ($Q = 4.0$), routing 3,738 slices (95.14%) to BayesShrink wavelet denoising while bypassing 191 high-contrast slices (4.86%). (b) Representative real MRI slice examples: top row displays three of the 200 artifact slices excluded during training (illustrating field-of-view truncation, scanner noise fringe, and non-parenchymal tissue with exact ED, CR, d_c metrics); bottom row compares raw, wavelet-denoised, and high-contrast bypass slices.

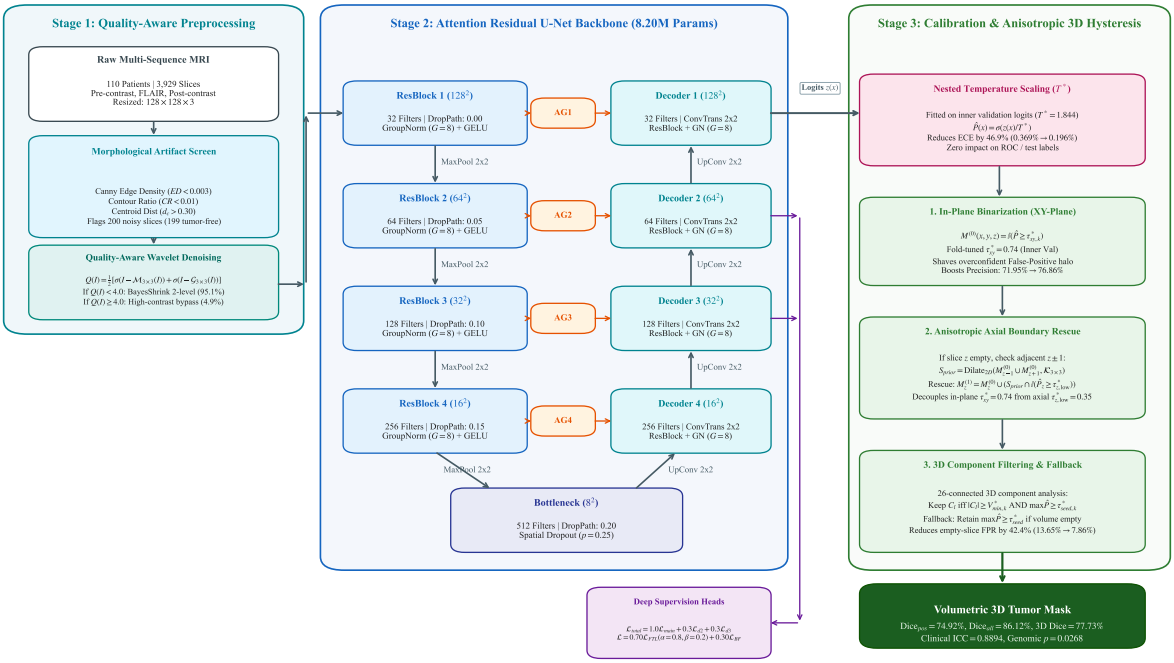


Figure 2: End-to-end CAL-3DH processing pipeline showing Quality-Aware Preprocessing (Stage 1), the Attention Res-UNet backbone with Group Normalization ($G = 8$), GELU, Stochastic Depth, and Deep Supervision (Stage 2), followed by Nested Temperature Scaling ($T^* = 1.844$) and Anisotropic 3D Z-Continuity Hysteresis (Stage 3).

3. Network Architecture and Nested Evaluation Protocol

3.1. Attention Residual U-Net Architecture

The segmentation backbone of CAL-3DH employs an Attention Residual U-Net (Attn Res-UNet) with deep supervision (Figure 2). Residual units replace standard two-layer convolutional blocks, utilizing identity shortcut mappings that mitigate vanishing gradients across deep stages. Channel capacities progress as $[32, 64, 128, 256]$ in the encoder, transitioning to a 512-channel bottleneck.

To ensure numerical stability when training with small mini-batch sizes ($B = 16$), all batch normalization layers are replaced with Group Normalization (GN) (Wu and He, 2018) using $G = 8$ groups:

$$x_{\text{norm}} = \gamma \cdot \frac{x - \mu_G}{\sqrt{\sigma_G^2 + \epsilon}} + \beta \tag{5}$$

where μ_G and σ_G are computed independently across spatial coordinates and group channels. Feature non-linearities utilize Gaussian Error Linear Units (GELU) (Hendrycks and Gimpel, 2016):

$$\text{GELU}(x) = x \cdot \Phi(x) = x \cdot P(X \leq x), \quad X \sim \mathcal{N}(0, 1) \tag{6}$$

Regularization is reinforced via Stochastic Depth (DropPath) (Huang et al., 2016) across residual branches:

$$\begin{aligned} H &= \text{GELU}\left(\text{shortcut}(x) + \frac{b}{1-p}\mathcal{F}(x)\right), \\ b &\sim \text{Bernoulli}(1-p) \end{aligned} \tag{7}$$

with linear decay rates $p \in [0.00, 0.05, 0.10, 0.15]$ in the encoder stages and $p = 0.20$ at the bottleneck.

In the decoder, additive attention gates (Oktay et al., 2018) dynamically recalibrate intermediate feature representations before concatenation. Given spatial features x^l and gating signal g, attention coefficients $\alpha^l \in [0, 1]$ are obtained via:

$$\alpha^l = \sigma\left(\psi^T\left(\text{GELU}\left(W_x^T x^l + W_g^T g + b_g\right)\right) + b_\psi\right) \tag{8}$$

where linear projections W_x and W_g align channel dimensions, and σ denotes the sigmoid activation. This gating suppresses background noise in peripheral brain regions while amplifying infiltrative tumor signals. Deep supervision auxiliary heads at decoders 2 and 3 provide intermediate gradients during backpropagation. The final network comprises exactly 8.20 million trainable parameters (8,202,005 parameters).

3.2. Loss Function and Training Dynamics

To address acute class imbalance (positive lesion pixels comprise less than 1.5% of total voxels), networks were trained using a compound production loss combining Focal Tversky Loss (FTL) (Salehi et al., 2017; Abraham and Khan, 2019) and Binary Focal Loss (BFL) (Lin et al., 2017):

$$\mathcal{L}_{\text{total}} = 0.70\,\mathcal{L}_{\text{FTL}} + 0.30\,\mathcal{L}_{\text{BF}} \tag{9}$$

The Focal Tversky formulation is defined as:

$$\mathrm{TI}(y, \hat{p}) = \frac{\sum_i y_i \hat{p}_i + \epsilon}{\sum_i y_i \hat{p}_i + \alpha \mathrm{FN}(y, \hat{p}) + \beta \mathrm{FP}(y, \hat{p}) + \epsilon} \tag{10}$$

$$\mathcal{L}_{\mathrm{FTL}} = (1 - \mathrm{TI})^{\gamma} \tag{11}$$

where $\gamma = 0.60$, $\epsilon = 1.0$, and the voxel-level error components are defined as $\mathrm{FN}(y, \hat{p}) = \sum_i y_i(1 - \hat{p}_i)$ and $\mathrm{FP}(y, \hat{p}) = \sum_i (1 - y_i)\hat{p}_i$.

In this formulation, $\alpha = 0.80$ is the penalty weight assigned to **False Negatives (FN)**, while $\beta = 0.20$ weights **False Positives (FP)**. Heavily penalizing false negatives ($\alpha = 0.80 \gg \beta = 0.20$) prioritizes the detection of diffuse, low-contrast tumor boundaries, achieving high baseline Recall (82.98%). However, this asymmetric penalty leads raw network probabilities to exhibit an intermediate-confidence false-positive halo surrounding lesions (raw Precision = 71.95%) and substantial probability miscalibration ($T^* = 1.844$). As detailed in Section 4, this peripheral false-positive halo is systematically eliminated in the CAL-3DH framework through nested calibration and anisotropic 3D Z-hysteresis filtering, elevating final Precision to 76.86%.

The Binary Focal Loss component is configured with focusing parameter $\gamma_{\mathrm{BF}} = 2.0$ and label smoothing $\epsilon_{\mathrm{LS}} = 0.02$. Auxiliary heads participate via multi-scale deep supervision:

$$\mathcal{L}_{\mathrm{multi}} = 1.0\, \mathcal{L}_{\mathrm{total}}^{(\mathrm{main})} + 0.30\, \mathcal{L}_{\mathrm{total}}^{(d2)} + 0.30\, \mathcal{L}_{\mathrm{total}}^{(d3)} \tag{12}$$

Optimization utilized the AdamW optimizer (Loshchilov and Hutter, 2019) with weight decay (10^{-4}) and a Cosine Annealing learning rate schedule featuring an 8-epoch linear warmup up to $\eta_{\max} = 3 \times 10^{-4}$, decaying to $\eta_{\min} = 10^{-6}$ over 100 epochs with early stopping patience of 20 epochs. Mini-batch size was fixed to 16. Stochastic data augmentation (random rotation $\pm 15°$, scaling 0.90–1.10, and elastic deformation with $\alpha = 15, \sigma = 3$) was active strictly during training. Horizontal flipping was intentionally excluded to preserve biological hemispheric asymmetry of gliomas. Validation and testing were performed in pure deterministic inference mode. All experiments were implemented in Python 3.11 with PyTorch 2.4 and CUDA 12.2 on an NVIDIA RTX 4090 GPU (24GB VRAM). A fixed global random seed (42) was enforced across all patient-grouped folds, weight initialization, and stochastic data augmentations to support computational reproducibility.

3.3. Patient-Grouped 5-Fold Cross-Validation

To prevent slice-level data leakage, 110 patients were partitioned into 5 balanced folds using stratified group splitting. All slices from any given patient were assigned strictly to a single fold.

In each fold $k \in \{1, \ldots, 5\}$, exactly 22 patients (20% of the 110-subject cohort, ≈ 786 slices per fold) were set aside as an untouched outer test partition (`outer_test`), strictly blinded and untouched during model training and tuning, and used solely to evaluate pre-specified pipeline configurations. The remaining 88 non-test patients were systematically partitioned in an 85%/15% ratio into 75 training patients ($\approx 2,535$ clean slices per fold

following artifact exclusion) utilized to optimize network parameters θ_k, and an internal holdout partition (`inner_val`) of 13 patients (467 slices per fold, 2,338 slices pooled across the 5 folds). This 13-patient `inner_val` split serves as the independent validation split for early-stopping checkpoint selection, scalar temperature scaling (T_k^*), and hysteresis hyperparameter selection ($\tau_{xy,k}^*, \tau_{z,\text{low},k}^*, V_{\min,k}^*, \tau_{\text{seed},k}^*$). While reserving 13 patients specifically for inner calibration reduces the training pool from 88 to 75 subjects, this design is structured to prevent any hyperparameter, temperature (T^*), or threshold (τ^*) leakage into the 22 held-out test patients.

Every test slice was evaluated strictly by the single dedicated model $\mathcal{M}_k(\cdot;\theta_k^*)$ trained for that fold:

$$\hat{Y}_i = \mathcal{M}_k(X_i;\theta_k^*), \quad \forall i \in \mathcal{D}_{\text{test}}^{(k)} \tag{13}$$

This 1-to-1 patient-to-fold mapping eliminates ensemble averaging ambiguity: no soft blending or multi-model averaging was utilized.

To quantify the precise degree of optimistic bias induced by slice-level data leakage, a matched parallel 5-fold Random-Slice cross-validation control experiment was conducted concurrently. Under this control protocol, the 3,929 axial slices were randomly partitioned into five equal folds without patient identity grouping, allowing slices from all 110 patients (100% patient overlap) to co-exist simultaneously across training, validation, and testing sets. Identical network architectures, learning rate schedules, loss formulations ($\alpha = 0.8, \beta = 0.2, \gamma = 0.6$), and stopping criteria were strictly enforced to isolate the effect of patient-level data leakage from model capacity.

3.4. Slice-Level Metric Definitions

In MRI volumes containing large proportions of healthy slices, evaluation metrics must handle non-tumor slices without distortion. We adhere to the standard clinical convention:

$$\text{Dice}(y,\hat{y}) = \begin{cases} 1.0, & \text{if } |y| = 0 \ \wedge \ |\hat{y}| = 0 \text{ (TN)} \\ 0.0, & \text{if } |y| = 0 \ \wedge \ |\hat{y}| > 0 \text{ (FP)} \\ \frac{2|y \cap \hat{y}|}{|y| + |\hat{y}|}, & \text{if } |y| > 0 \end{cases} \tag{14}$$

Three primary slice-level metrics are reported to comprehensively capture clinical utility: Dice_{pos}, calculated strictly over the 1,373 tumor-bearing slices ($|y| > 0$); Dice_{all}, computed over all 3,929 slices including healthy non-tumor tissue; and the empty-slice false-positive rate ($\text{FPR}_{\text{empty}}$), quantifying the percentage of healthy slices erroneously predicted to contain lesion voxels ($|\hat{y}| > 0$ when $|y| = 0$).

4. Calibration and Anisotropic 3D Z-Hysteresis (CAL-3DH)

4.1. Nested Temperature Scaling Calibration

Because overlap losses yield uncalibrated probability estimates, raw network logits $z(x)$ were calibrated via post-processing temperature scaling (Guo et al., 2017). Temperature

parameter $T_k > 0$ was optimized strictly on inner validation logits by minimizing negative log-likelihood (NLL):

$$\hat{T}_k^* = \arg\min_{T>0} \sum_{j \in \mathcal{D}_{\text{val}}^{(k)}} \Big[-y_j \log \sigma(z_j/T) - (1-y_j) \log\big(1 - \sigma(z_j/T)\big) \Big] \tag{15}$$

Because temperature scaling applies a monotonic scalar transformation ($T_k > 0$), it preserves the relative ranking of logits while softening extreme confidence values. Expected Calibration Error (ECE) was calculated across $M = 15$ equally spaced probability bins:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right| \tag{16}$$

To quantify miscalibration within the clinically critical infiltrative margin region while preventing background dilution by easy non-tumor voxels, we define the active-bin calibration (overconfidence) gap across intermediate-probability bins $\mathcal{M}_{\text{active}} = \{m \mid \text{bin}_m \subseteq [0.2, 0.9]\}$, where $\text{conf}(B_m) > \text{acc}(B_m)$ across all active bins in Figure 3(a):

$$\text{Gap}_{\text{active}} = \frac{1}{|\mathcal{M}_{\text{active}}|} \sum_{m \in \mathcal{M}_{\text{active}}} \left(\text{conf}(B_m) - \text{acc}(B_m)\right) \tag{17}$$

and evaluate the corresponding voxel-weighted active-bin Expected Calibration Error:

$$\text{ECE}_{\text{active}} = \frac{\sum_{m \in \mathcal{M}_{\text{active}}} |B_m| \left|\text{acc}(B_m) - \text{conf}(B_m)\right|}{\sum_{m \in \mathcal{M}_{\text{active}}} |B_m|} \tag{18}$$

which directly measures infiltrative margin miscalibration and corresponds to the 36.14% to 29.57% reduction reported in Section 5.2.

The functional role of calibration in the CAL-3DH pipeline is fundamental to multi-parameter volumetric post-processing. While a single in-plane threshold $\tau_{xy,k}^*$ could be mapped monotonically to an uncalibrated threshold $\tau_{\text{raw},k}^* = \sigma(T_k^* \, \text{logit}(\tau_{xy,k}^*))$, CAL-3DH evaluates candidate 3D hysteresis grids ($\tau_{z,\text{low}} \in \{0.35, 0.40, 0.45, 0.50\}$, $\tau_{\text{seed}} \in \{0.70, 0.75\}$) across folds whose uncalibrated logit scales vary substantially ($T_k^* \in [1.520, 2.387]$); calibrating probabilities first aligns all folds onto a common probability scale and provides better-calibrated voxel-level probability estimates along ambiguous infiltrative margins, ensuring robust generalization of downstream spatial rules across unseen test volumes.

4.2. Anisotropic 3D Z-Continuity Hysteresis Algorithm

In 2D axial segmentation, high decision thresholds suppress false alarms but truncate apical and basal tumor margins, whereas low thresholds preserve boundaries at the expense of false alarms on healthy slices. CAL-3DH resolves this trade-off through an anisotropic 3D Z-continuity hysteresis algorithm operating across calibrated probabilities

$\hat{P} = \sigma(z/\hat{T}_k^*)$. All post-processing hyperparameters were optimized strictly on internal validation splits (`inner_val`, 13 patients): (1) scalar temperature $T_k \in [0.5, 4.0]$ via bounded scalar minimization of validation NLL; (2) planar threshold $\tau_{xy,k} \in [0.20, 0.90]$ with step 0.02 maximizing validation Dice_{pos}; and (3) 3D spatial parameters via grid search over $\tau_{z,\text{low}} \in \{0.35, 0.40, 0.45, 0.50\}$, $V_{\min} \in \{20, 50, 80\}$, and $\tau_{\text{seed}} \in \{0.70, 0.75\}$ maximizing inner-validation slice-averaged positive Dice (Dice_{pos}):

1. **In-Plane Planar Binarization:** Each slice is initially thresholded with fold-specific inner-validation threshold $\tau_{xy,k}^*$ ($0.62 \leq \tau_{xy}^* \leq 0.84$, mean: 0.74):
$$M^{(0)}(x, y, z) = \mathbb{I}\left(\hat{P}(x, y, z) \geq \tau_{xy,k}^*\right) \tag{19}$$
preserving horizontal tumor boundary fidelity and shaving the FP halo.
2. **Z-Continuity Boundary Rescue:** For slices where $M^{(0)}$ detects zero (or near-zero, < 10 pixels) tumor tissue, adjacent slices $z-1$ and $z+1$ are inspected. If an adjacent slice contains a verified tumor (≥ 20 pixels), a dilated spatial footprint $\mathcal{S}_{\text{prior}}$ is projected onto slice z:
$$\mathcal{S}_{\text{prior}} = \text{Dilate}_{2D}\left(M_{z-1}^{(0)} \cup M_{z+1}^{(0)}, \mathcal{K}_{3\times3}\right) \tag{20}$$
Voxels within $\mathcal{S}_{\text{prior}}$ satisfying a relaxed axial threshold $\tau_{z,\text{low},k}^*$ (mean $\tau_{z,\text{low}}^* = 0.35$, where $\tau_{z,\text{low}}^* < \tau_{xy}^*$) are rescued:
$$M_z^{(1)} = M_z^{(0)} \cup \left(\mathcal{S}_{\text{prior}} \cap \mathbb{I}(\hat{P}_z \geq \tau_{z,\text{low},k}^*)\right) \tag{21}$$
3. **3D Volumetric Suppression:** On the resulting 3D volume, 26-connected components $\{C_l\}$ are labeled. Components are discarded if their 3D volume is smaller than $V_{\text{min},k}^*$ (mean $V_{\min}^* = 38$ voxels) or their peak probability is below $\tau_{\text{seed},k}^*$ (mean $\tau_{\text{seed}}^* = 0.70$):
$$\begin{aligned}\mathcal{C}_{\text{valid}} = \big\{C_l \,||C_l| \geq V_{min,k}^* \wedge \\ \max_{v \in C_l} \hat{P}(v) \geq \tau_{seed,k}^*\big\}\end{aligned} \tag{22}$$
4. **Fallback Safeguard:** To prevent small apical tumors from collapsing to empty volumes (which imposes maximal boundary distance penalties), if $\mathcal{C}_{\text{valid}} = \emptyset$, components with $\max \hat{P} \geq \tau_{\text{seed}}^*$ are retained. This safeguard is active in both the 3D-filtered ablation (Step 1 + Step 3) and Full CAL-3DH.

The candidate search lower bounds ($\tau_{z,\text{low}} \geq 0.35$, $\tau_{\text{seed}} \geq 0.70$) were grounded in explicit biophysical and calibration constraints: lowering $\tau_{z,\text{low}}$ below 0.35 encroaches upon the low-confidence background noise floor and triggers excessive false-positive bleeding into adjacent tumor-free slices (empirically verified on internal validation splits, where relaxing $\tau_{z,\text{low}}$ to 0.25–0.30 degrades specificity by inflating empty-slice false alarms without rescuing additional true boundary slices), whereas lowering the 3D component seed threshold τ_{seed} below 0.70 drops below the calibrated in-plane decision plateau ($\tau_{xy}^* \in [0.62, 0.84]$, mean 0.74), violating the hysteresis requirement that every retained 3D lesion component must be anchored by at least one high-confidence core voxel.

Table 3: Baseline out-of-fold performance at fixed $\tau = 0.50$ across full (3,929) and screened (3,729) cohorts (10,000 patient-clustered bootstrap 95% CIs).

Evaluation Metric	Full Cohort (3,929)	Screened Cohort (3,729)
Dice_{pos} (%)	73.96 [70.10 – 77.15]	73.96 [70.10 – 77.13]
Dice_{all} (%)	82.02 [79.75 – 84.10]	82.21 [80.01 – 84.27]
IoU_{pos} (%)	63.97 [59.78 – 67.89]	63.97 [59.78 – 67.88]
$\text{Precision}_{\text{pos}}$ (%)	71.95 [67.80 – 75.80]	71.95 [67.80 – 75.80]
$\text{Recall}_{\text{pos}}$ (%)	82.98 [78.60 – 86.70]	82.98 [78.60 – 86.70]
$\text{FPR}_{\text{empty}}$ (%)	13.65 (349 / 2556)	13.41 (316 / 2357)

5. Experimental Results and Leakage Auditing

5.1. Baseline Segmentation Performance

Table 3 summarizes baseline Attn Res-UNet performance across all 3,929 out-of-fold test slices at fixed threshold $\tau = 0.50$. Ninety-five percent confidence intervals were computed via 10,000 patient-clustered bootstrap iterations (Efron and Tibshirani, 1994). The model achieved a positive-slice Dice of 73.96% [95% CI: 70.10%–77.15%], an overall Dice of 82.02% [95% CI: 79.75%–84.10%], an intersection-over-union (IoU_{pos}) of 63.97%, and an empty-slice false-positive rate of 13.65%.

5.2. Nested Temperature Scaling and Thresholding

Table 4 presents fold-by-fold calibration and threshold optimization results. Fitted optimal temperatures ranged from $T^* = 1.520$ to 2.387 (mean $T^* = 1.844$). On untouched test sets, temperature scaling consistently reduced NLL from 0.0230 to 0.0154 (-32.9%) and Brier score (Brier, 1950) from 0.00375 to 0.00338 (-9.9%). The mean per-fold ECE decreased from 0.372% to 0.254% (a 31.7% relative reduction), while pooled cohort ECE dropped from 0.369% to 0.196% (a 46.9% relative reduction), visually confirmed in Figure 3.

Crucially, the small absolute magnitude of cohort ECE (0.369%) stems from voxel dilution: over 98.5% of MRI voxels belong to background air and healthy, non-enhancing brain tissue ($y = 0, \hat{p} \approx 0$, clearly reflected in the first bin of the logarithmic histogram in Figure 3(b) encompassing $> 6 \times 10^7$ voxels). Within the clinically vital active probability bins corresponding to the ambiguous infiltrative tumor margins ($\hat{p} \in [0.2, 0.9]$ in Figure 3(a)), the raw uncalibrated network exhibits severe overconfidence, manifested by an Overconfidence Gap of 0.253 (25.3%, Section 5.5). Nested temperature scaling ($T^* = 1.844$) contracts this margin overconfidence gap down to 0.200 (a 20.9% relative reduction), substantially shifting empirical confidence toward the ideal diagonal while active-margin weighted ECE decreases from 36.14% to 29.57%. The corresponding training and validation convergence dynamics across folds are illustrated in Figure 4. In Figure 4(a), validation loss is slightly lower than training loss during early epochs because heavy stochastic data augmentation, DropPath stochastic depth ($p \leq 0.20$), and label smoothing ($\epsilon_{\text{LS}} = 0.02$) act as strong regularizers during training and are deactivated during deterministic validation; in contrast,

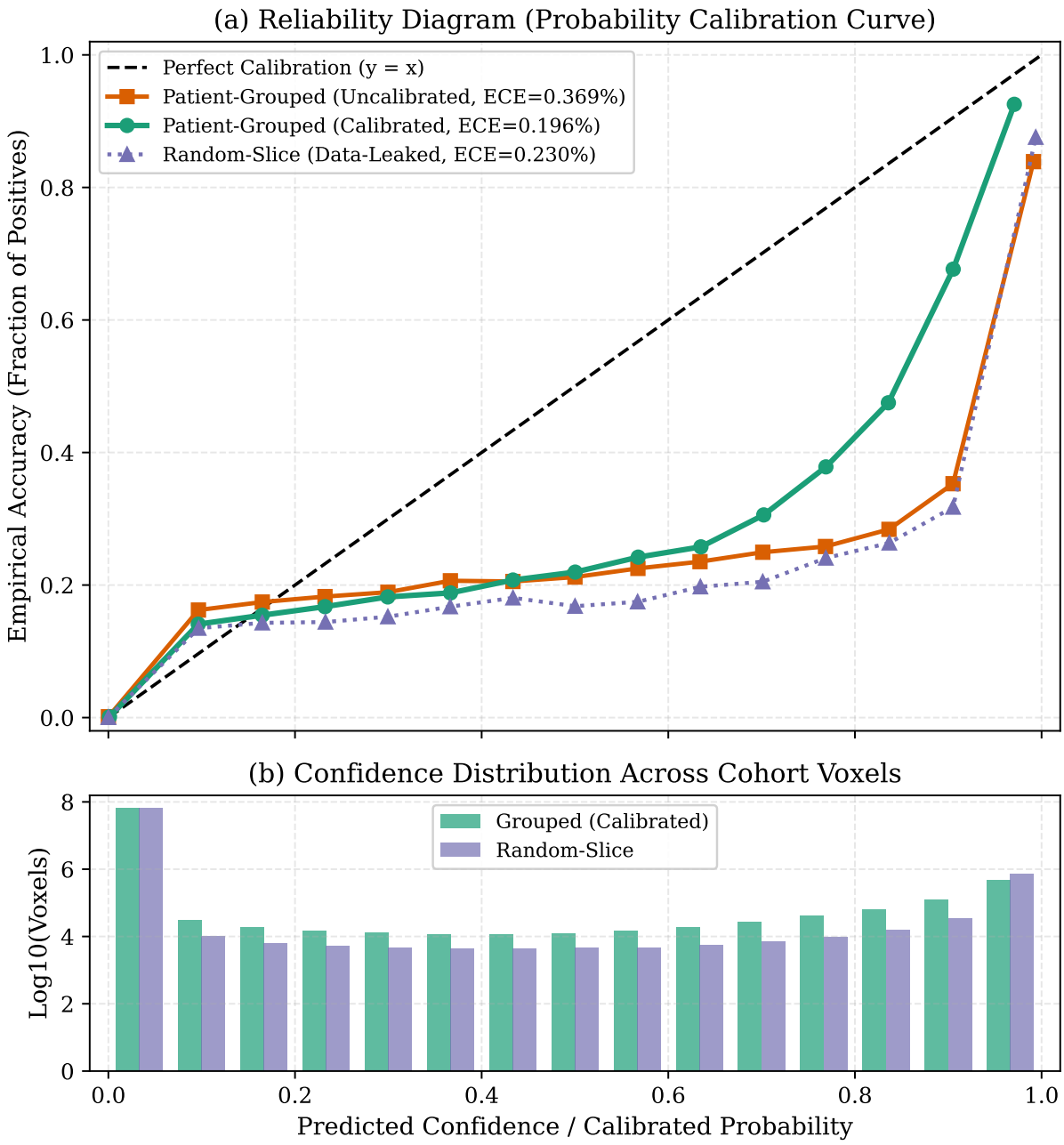


Figure 3: Probability calibration and voxel confidence dynamics across 15 probability bins. (a) Reliability diagram comparing the uncalibrated patient-grouped model (ECE = 0.369%), the temperature-calibrated model ($T^* = 1.844$, reducing pooled ECE by 46.9% to 0.196%), and the data-leaked random-slice model against ideal diagonal calibration. (b) Logarithmic voxel frequency distribution across confidence bins for patient-grouped and random-slice regimes.

Table 4: Fold-by-fold nested calibration, threshold optimization, and out-of-fold performance.

Fold	Optimal $\mathbf{T}^*$	Inner $\tau^*_{\mathbf{xy}}$	Inner $\tau^*_{\mathbf{z},\text{low}}$	Inner $\mathbf{V}^*_{\text{min}}$	Inner τ^*_{seed}	Outer ECE (%)		Outer NLL		Outer Test Dice_{pos} (%)		
						Raw	Calibrated	Raw	Calibrated	2D Base ($\tau = 0.50$)	2D Nested (τ^*_{xy})	CAL-3DH (Ours)
1	2.019	0.76	0.35	50	0.70	0.354	0.153	0.0267	0.0159	77.02	77.00	**77.26**
2	2.387	0.62	0.35	20	0.70	0.330	0.227	0.0191	0.0112	74.96	75.01	**75.04**
3	1.649	0.82	0.35	80	0.70	0.391	0.367	0.0186	0.0141	79.39	81.91	**82.39**
4	1.520	0.84	0.35	20	0.70	0.476	0.335	0.0325	0.0232	70.44	70.50	**70.96**
5	1.644	0.68	0.35	20	0.70	0.308	0.186	0.0181	0.0127	68.15	68.61	**69.01**
Mean	**1.844**	**0.74**	**0.35**	**38**	**0.70**	**0.372**	**0.254**	**0.0230**	**0.0154**	**73.96**	**74.60**	**74.92**

in the random-slice regime (Figure 4(c)–(d)), validation Dice unnaturally surpasses training Dice by nearly 15 pp due to subject memorization across contiguous slices.

5.3. Quantification of Optimism Bias (ΔOptimism)

To rigorously disentangle sources of performance overestimation, we quantified the empirical difference between nested inner-fold thresholding and pooled test threshold sweeping (Figure 5). When decision thresholds were selected strictly on independent inner validation partitions (mean calibrated $\tau^*_{xy} = 0.74$, corresponding to raw uncalibrated threshold 0.86), the model achieved an out-of-fold test Dice_{pos} of 74.60% [95% CI: 70.45%–78.08%]. Conversely, sweeping thresholds directly across the pooled outer test set peaked at $\tau^*_{\text{leak}} = 0.76$ (raw threshold 0.88), yielding an apparent Dice_{pos} of 74.66%. The resulting empirical optimism bias is:

$$\Delta\text{Optimism} = 74.66\% - 74.60\% = +0.063 \text{ pp} \tag{23}$$

This modest optimism bias (+0.063 pp) contrasts sharply with the massive +8.79 pp inflation generated by slice-level data leakage (Section 5.5). The primary engine of artificial performance inflation in prior literature is subject identity memorization across contiguous slices, whereas nested inner-fold thresholding represents an essential methodological best practice. Because the Dice plateau is remarkably stable across calibrated probabilities $\hat{P} \in [0.68, 0.85]$ (Figure 5), the empirical risk of post-hoc threshold tuning is negligible. This plateau stability indicates that transitioning from the default, suboptimal threshold ($\tau = 0.50$) to the nested optimal threshold ($\tau^*_{xy} = 0.74$) legitimately captures +0.64 pp in Dice_{pos} (73.96% to 74.60%), a +2.43 pp jump in overall Dice_{all} (82.02% to 84.45%), and a 24.9% relative reduction in empty-slice false alarms (349 to 262 slices) with zero test partition snooping.

5.4. CAL-3DH Performance and Step-by-Step Anisotropic 3D Z-Hysteresis Ablation

In clinical volumetric FLAIR protocols where 65.05% of axial slices are entirely tumor-free (2,556/3,929 slices), the primary clinical advantage of CAL-3DH lies in eliminating **148 false alarm slices relative to the baseline (−42.4%, dropping from 349 to 201 slices)** and **61 fewer false alarm slices compared to 2D Nested (−23.3% additional relative reduction, from 262 to 201 slices)**. This selective pruning propels overall

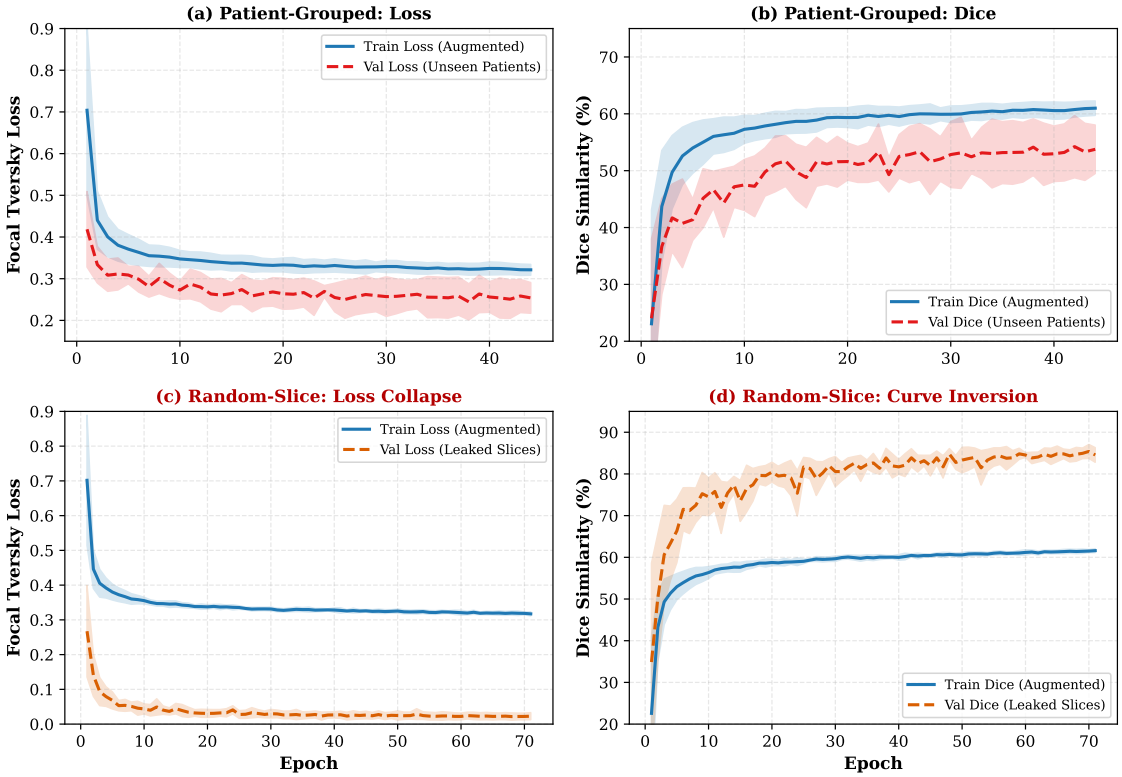


Figure 4: Training and validation convergence dynamics across 5 folds under identical training hyperparameters. (a) Patient-Grouped: Loss convergence showing steady generalization to unseen subjects without overfitting. (b) Patient-Grouped: Dice progression reaching stable plateau. (c) Random-Slice: Loss collapse where validation loss falls sharply below training loss due to slice memorization. (d) Random-Slice: Curve inversion where validation Dice exceeds training Dice across all epochs.

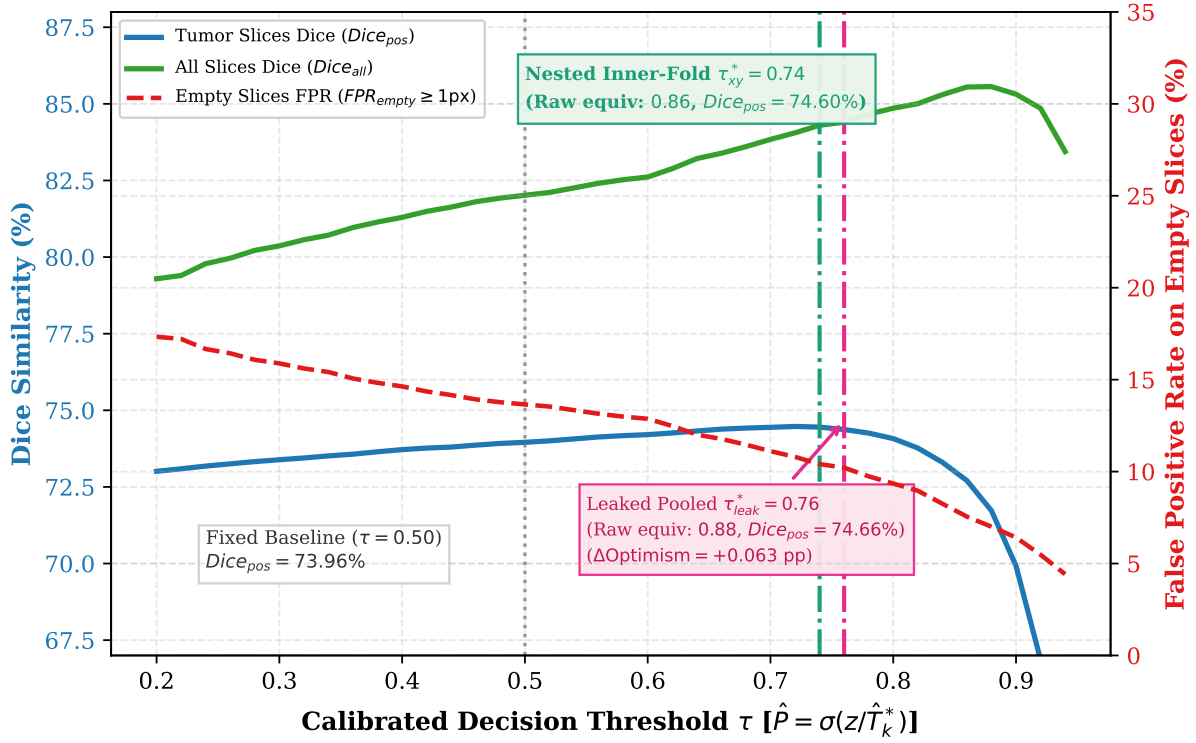


Figure 5: Decision threshold sensitivity analysis on out-of-fold predictions plotted across calibrated probabilities $\hat{P} = \sigma(z/\hat{T}^*_k)$. The curve illustrates the operational trade-off between tumor-slice Dice (Dice_{pos}), overall Dice (Dice_{all}), and false alarm rate on empty slices ($\text{FPR}_{\text{empty}}$). Vertical markers denote the standard baseline ($\tau = 0.50$), the nested inner-fold calibrated threshold ($\tau^*_{xy} = 0.74$, corresponding to raw uncalibrated threshold 0.86), and the post-hoc leaked threshold ($\tau^*_{\text{leak}} = 0.76$, raw equivalent 0.88).

volumetric accuracy (Dice_{all}) by +4.10 pp (+5.00% relative gain) over baseline and +1.67 pp over 2D Nested ($p < 0.0001$, patient-level paired Wilcoxon signed-rank test).

Crucially, unlike conventional isotropic 3D volumetric filters that cause axial-transverse blurring and degrade horizontal in-plane sensitivity on clinical 2D axial brain MRI acquisitions (where physical slice thickness and through-plane z-spacing are typically 4–6 mm vs. ∼1 mm in-plane prior to raster export), CAL-3DH preserves high-fidelity planar boundaries while recovering discontiguous apical and basal slices via confident 2D spatial seeds. Consequently, positive-slice Dice reaches 74.92% [95% CI: 70.82%–78.39%], uniformly improving across all five cross-validation folds simultaneously (Table 4).

To disentangle the specific biophysical contributions of each algorithmic component in Stage 3, Table 5 details the step-by-step ablation across all 3,929 out-of-fold test slices:

1. **Step 1 Only (2D Nested Thresholding, τ^*_{xy}):** Raising the in-plane decision threshold from 0.50 to the calibrated inner-fold optimum $\tau^*_{xy} = 0.74$ elevates Dice_{pos} from 73.96% to 74.60% and improves Precision from 71.95% to 76.51%, reducing empty-slice false alarms by 24.9% (349 to 262 slices). However, it leaves missed apical/basal boundary slices unaddressed (0/220 rescued).
2. **Step 1 + Step 2 (+ Axial Z-Continuity Rescue, $\tau^*_{z,\text{low}}$):** Activating axial continuity rescue ($\tau^*_{z,\text{low}} = 0.35$ on slices adjacent to confident 2D seeds) boosts Recall from 80.14% to 80.62%, elevates boundary slice Dice from 29.85% to 30.39%, and successfully rescues 4 previously completely missed apical boundary slices (4/220, 1.82%). Because low-threshold propagation can expand into adjacent slices, empty-slice false alarms slightly increase to 291 slices (11.38%) in the absence of volume filtering.
3. **Step 1 + Step 3 (+ 3D Component Filtering, $V^*_{\text{min}}, \tau^*_{\text{seed}}$):** Enforcing 3D connected-component volume filtering ($V^*_{\text{min}} = 38$ voxels, $\tau^*_{\text{seed}} = 0.70$) without Z-rescue aggressively suppresses spurious noise, dropping empty-slice false alarms to 176 slices (6.89%) and increasing Dice_{all} to 86.66% and Precision to 77.34%. However, without Z-rescue, missed boundary slices remain unrecovered (0/220).
4. **Full CAL-3DH (Step 1 + Step 2 + Step 3 + Fallback):** Combining both steps harmonizes the operational sensitivity-specificity trade-off through a complementary dual mechanism. Relative to **Step 1 + Step 2** (axial Z-rescue alone, Row 3: $\text{Dice}_{\text{pos}} = 74.88\%$, $\text{Dice}_{\text{all}} = 83.81\%$, $\text{FPR}_{\text{empty}} = 11.38\%$), adding 3D connected-component volume filtering (Step 3) retains all 4 rescued apical boundary slices while yielding a modest +0.04 pp gain in Dice_{pos} (74.88% to 74.92%) and +0.09 pp in boundary Dice (30.39% to 30.48%), while crucially eliminating **90 false-alarm slices** (291 → 201 slices; a −3.52 pp or −30.9% relative reduction in $\text{FPR}_{\text{empty}}$ from 11.38% to 7.86%) and boosting overall volumetric accuracy (Dice_{all}) by +2.31 pp (83.81% to 86.12%), justifying Step 3 primarily on volumetric specificity grounds. Conversely, relative to **Step 1 + Step 3** (3D filtering alone, Row 4: $\text{Dice}_{\text{pos}} = 74.64\%$, $\text{Dice}_{\text{all}} = 86.66\%$, $\text{FPR}_{\text{empty}} = 6.89\%$), incorporating axial Z-continuity rescue (Step 2) improves positive-slice Dice by +0.28 pp (74.64% to 74.92%) and boundary-slice Dice by +0.63 pp (29.85% to 30.48%, rescuing 4/220 missed apical slices) at the modest cost of 25 additional empty-slice false alarms (176 → 201 slices; $\text{FPR}_{\text{empty}} = 6.89\%$ to 7.86%).

Table 5: Step-by-step ablation of Stage 3 (Anisotropic 3D Z-Hysteresis) across all 3,929 out-of-fold test slices ($n = 110$ patients). Boundary slices ($n = 220$) represent the first and last tumor-bearing axial slices adjacent to healthy tissue across all 110 volumes (2×110).

Configuration / Ablation Stage	Dice_{pos} (%)	Dice_{all} (%)	**Precision (%)**	**Recall (%)**	$\text{FPR}_{\text{empty}}$ % (count)	**Boundary Dice (%)**	**Rescued (/220)**
(1) 2D Baseline ($\tau = 0.50$)	73.96	82.02	71.95	**82.98**	13.65 (349 / 2556)	29.68	0 / 220
(2) Step 1: 2D Nested ($\tau^*_{xy} = 0.74$)	74.60	84.45	76.51	80.14	10.25 (262 / 2556)	29.85	0 / 220
(3) Step 1 + Step 2 (+ Axial Z-Rescue, $\tau^*_{z,\text{low}}$)	74.88	83.81	75.82	80.62	11.38 (291 / 2556)	30.39	**4 / 220**
(4) Step 1 + Step 3 (+ 3D Filtering, $V^*_{\text{min}}, \tau^*_{\text{seed}}$)	74.64	**86.66**	**77.34**	79.45	**6.89 (176 / 2556)**	29.85	0 / 220
(5) CAL-3DH (Full) (Step 1 + 2 + 3 + Fallback)	**74.92**	86.12	76.86	79.74	7.86 (201 / 2556)	**30.48**	**4 / 220**

Patient-Level Paired Testing ($n = 110$ subjects): CAL-3DH vs. Baseline: Dice_{all} $W = 668, p < 0.0001$; $\text{FPR}_{\text{empty}}$ $W = 75.5, p < 0.0001$; 3D Volumetric Dice $W = 927, p < 0.0001$. CAL-3DH vs. 2D Nested: Dice_{all} $W = 687, p < 0.0001$; $\text{FPR}_{\text{empty}}$ $W = 240, p = 0.0002$; 3D Volumetric Dice $W = 645, p < 0.0001$ (paired Wilcoxon signed-rank tests).

Because our primary pre-specified clinical objective prioritizes tumor-bearing sensitivity and positive-slice overlap (Dice_{pos}), Full CAL-3DH is selected as the primary clinical configuration while still achieving a substantial 42.4% relative false-alarm reduction over the 2D baseline.

To account for inter-slice clustering within patients ($n = 110$), we evaluated valid patient-level paired Wilcoxon signed-rank tests alongside 10,000-resample paired bootstrap 95% confidence intervals of the patient-level mean differences:

- **Patient-Averaged Dice_{all}:** CAL-3DH achieves $86.16 \pm 8.78\%$ [95% CI: 84.53%–87.75%], significantly outperforming 2D Baseline ($82.15 \pm 10.58\%$, $W = 668, p < 0.0001$; paired mean difference $\Delta\text{Dice}_{\text{all}} = +4.01$ pp, 95% CI: $[+2.91, +5.24]$ pp) and 2D Nested ($84.48 \pm 9.33\%$, $W = 687, p < 0.0001$; paired mean difference $\Delta\text{Dice}_{\text{all}} = +1.68$ pp, 95% CI: $[+0.98, +2.45]$ pp).

- **Patient Empty-Slice False Alarms ($\text{FPR}_{\text{empty}}$):** CAL-3DH suppresses false alarms to $7.68 \pm 10.37\%$ [95% CI: 5.81%–9.67%], demonstrating highly significant reductions compared to 2D Baseline ($13.25 \pm 14.05\%$, $W = 75.5, p < 0.0001$; paired mean difference $\Delta\text{FPR}_{\text{empty}} = -5.57$ pp, 95% CI: $[-7.38, -3.99]$ pp) and 2D Nested ($10.09 \pm 11.45\%$, $W = 240, p = 0.0002$; paired mean difference $\Delta\text{FPR}_{\text{empty}} = -2.41$ pp, 95% CI: $[-3.69, -1.30]$ pp).

- **Patient 3D Volumetric Dice:** As shown in Table 7, CAL-3DH yields $77.73 \pm 18.65\%$ [95% CI: 74.20%–81.18%], achieving statistically significant superiority over 2D Baseline ($76.00 \pm 17.94\%$, $W = 927, p < 0.0001$; paired mean difference Δ3D Dice $= +1.73$ pp, 95% CI: $[+1.18, +2.28]$ pp) and 2D Nested ($77.57 \pm 18.66\%$, $W = 645, p < 0.0001$; paired mean difference Δ3D Dice $= +0.16$ pp, 95% CI: $[+0.10, +0.23]$ pp).

Table 6: Systematic data leakage audit comparing patient-grouped against random-slice 5-fold cross-validation on the TCGA-LGG cohort (10,000-resample bootstrap 95% CIs in brackets).

Protocol	Leakage Status	Dice_{pos} (%)	Dice_{all} (%)	Pooled ECE (%)	Brier Score	Overconf. Gap
Patient-Grouped (Uncalibrated)	Zero Leakage	73.96 [70.10, 77.15]	82.02 [79.75, 84.10]	0.369	0.00375	0.253
Patient-Grouped (Calibrated T^*)	Zero Leakage	73.96 [70.10, 77.15]	82.02 [79.75, 84.10]	**0.196**	**0.00338**	**0.200**
Random-Slice (Conventional)	**Slice Memorization**	**82.74 [81.75, 83.72]**	**88.22 [87.40, 89.02]**	0.230	0.00228	0.278
Leakage Impact	Memorization Distortion	**+8.79 pp**	**+6.20 pp**	–	–	–
Calibration Impact	Uncertainty Regularization	0.00 pp	0.00 pp	**-46.9%**	**-9.9%**	**-20.9%**

5.5. Data Leakage Audit: Patient-Grouped vs. Random-Slice

Table 6 contrasts patient-grouped partitioning against random-slice partitioning under identical network architectures and loss formulations. As demonstrated by the empirical convergence dynamics (Figure 4(c)–(d)), random slice splitting leads to an unnatural loss collapse and curve inversion, where validation performance surpasses training performance because adjacent slices are shared across partitions.

Random slice splitting artificially inflated positive-slice Dice by **+8.79 pp** (+11.88% relative inflation, from 73.96% to 82.74%) and overall Dice by **+6.20 pp** (+7.56% relative inflation, from 82.02% to 88.22%). Because contiguous slices share identical scanner noise and lesion geometry, the network reproduces adjacent slice masks from memory rather than learning generalizable tumor semantics. Multi-scenario qualitative segmentation results across representative test slices are shown in Figure 6.

6. 3D Volumetric Agreement and Radiogenomic Fidelity

6.1. Patient-Level 3D Surface and Volumetric Assessment

Individual 2D predictions were stacked into full 3D volumes across all 110 subjects. Table 7 details 3D Volumetric Dice, 95% Hausdorff Distance (HD_{95}), and Average Surface Distance (ASD).

The proposed CAL-3DH framework consistently outperforms both 2D baselines across all patient-level volumetric metrics. 3D Volumetric Dice reaches 77.73%±18.65% with a median of 84.42% (interquartile range [IQR]: 70.85%–91.24%). Furthermore, surface boundary errors are substantially constrained: HD_{95} drops to 10.97 ± 18.28 voxels (median: 2.24, IQR: 1.00–12.03), while ASD improves to 3.10 ± 12.48 voxels (median: 0.63, IQR: 0.29–2.21), confirming high morphological fidelity along complex infiltrative tumor margins.

6.2. Clinical Agreement and Bland-Altman Analysis

Agreement between automated 3D volumes and expert manual ground truth was evaluated using standardized clinical biostatistical tests (Figure 7). Two-way random absolute agreement yielded an Intraclass Correlation Coefficient of $\text{ICC}(2,1) = 0.8894$ (95% CI: $[0.7782, 0.9521]$), indicating good-to-excellent clinical measurement reliability according to standard guidelines (Shrout and Fleiss, 1979; Koo and Li, 2016). Linear and monotonic

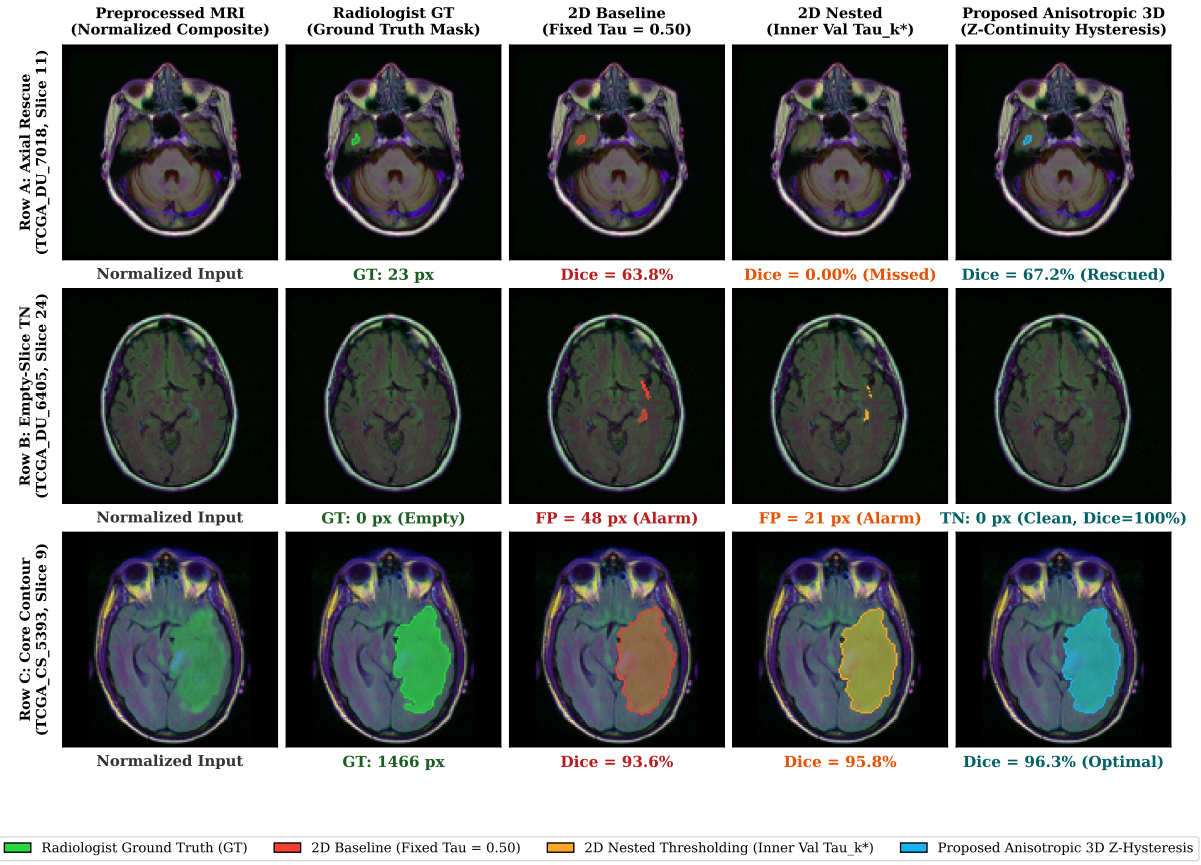


Figure 6: Multi-scenario qualitative segmentation comparison across representative test slices. Row A (Axial Rescue): Boundary slice rescue along the Z-axis, where 2D Nested misses apical tumor (Dice = 0.0%) while CAL-3DH restores detection (Dice = 67.2%). Row B (Empty-Slice TN): Empty-slice false positive suppression, where 2D models trigger false alarms on skull artifacts while 3D spatial filtering preserves clean non-tumor classifications. Row C (Core Contour): High geometric fidelity on large, complex infiltrative core lesions (Dice = 96.3%).

Table 7: Patient-level 3D volumetric and surface boundary metrics across all 110 TCGA-LGG patients. Pairwise comparisons of 3D Volumetric Dice confirm that CAL-3DH statistically significantly outperforms both 2D Baseline ($W = 927, p < 0.0001$) and 2D Nested ($W = 645, p < 0.0001$, paired Wilcoxon signed-rank test).

Method	3D Volumetric Dice (%)	HD_{95} (voxels)	ASD (voxels)	3D Prec. (%)	3D Rec. (%)
2D Baseline ($\tau = 0.50$)	76.00 ± 17.94 (Med: 82.96)	12.89 ± 19.21 (Med: 3.57)	3.59 ± 12.47 (Med: 0.88)	71.89	**85.96**
2D Nested (τ^*_{xy})	77.57 ± 18.66 (Med: 84.42)	11.26 ± 18.32 (Med: 2.34)	3.18 ± 12.48 (Med: 0.76)	77.40	82.98
CAL-3DH (Ours)	**77.73 ± 18.65 (Med: 84.42)**	**10.97 ± 18.28 (Med: 2.24)**	**3.10 ± 12.48 (Med: 0.63)**	**77.70**	82.99

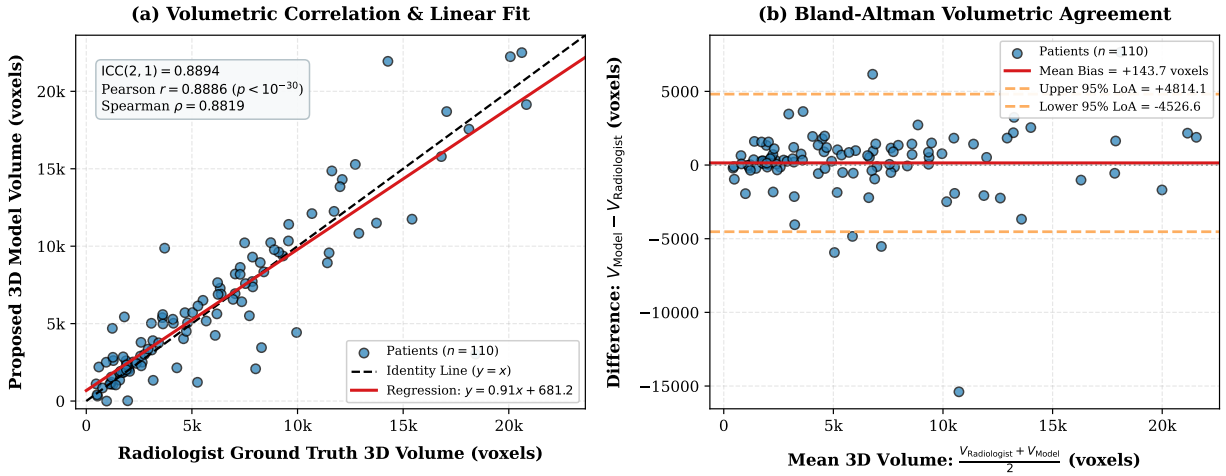


Figure 7: Clinical 3D volumetric agreement and calibration analysis across 110 patients. (a) Volumetric correlation and linear regression fit between automated 3D predictions and radiologist ground truth (ICC$(2,1) = 0.8894$, Pearson $r = 0.8886, p = 2.44 \times 10^{-38}$, Spearman $\rho = 0.8819$). (b) Bland-Altman agreement plot with minimal mean bias (+143.7 voxels, 2.38%, under 2.5% of mean lesion volume) and 95% limits of agreement $[-4{,}526.6, +4{,}814.1]$ voxels.

agreements were corroborated by Pearson correlation ($r = 0.8886, p = 2.44 \times 10^{-38}$) and Spearman rank correlation ($\rho = 0.8819, p = 4.72 \times 10^{-37}$). Bland-Altman agreement analysis (Bland and Altman, 1986) revealed a minimal mean volumetric bias of +143.7 voxels (accounting for 2.38%, under 2.5% of mean lesion volume ≈ 6,027 voxels), with 95% limits of agreement spanning $[-4{,}526.6, +4{,}814.1]$ voxels. As illustrated in Figure 7(b), these limits of agreement reflect heavy-tailed variance driven by a small subset of large, diffusely infiltrative grade III tumors (> 15,000 voxels), whereas bias remains tightly centered near zero for small and moderate lesion burdens.

6.3. Radiogenomic Stratification with Clinical Data

To verify whether automated segmentations support downstream clinical biomarker extraction, we evaluated whether predicted 3D geometries reproduce known biological associations documented in the TCGA pan-glioma atlas (Figure 8). Genomic and histological

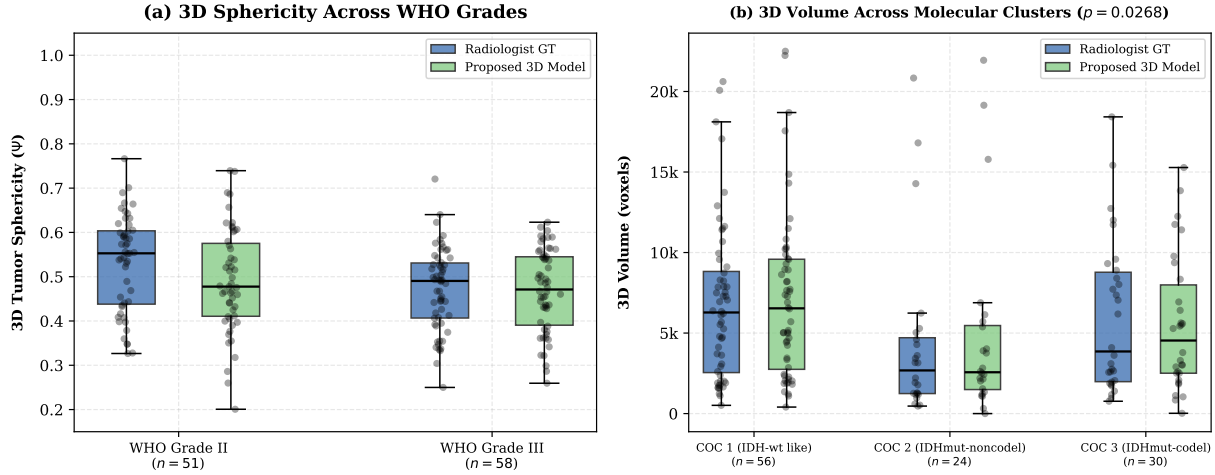


Figure 8: Downstream radiogenomic fidelity and histological stratification with clinical records. (a) 3D tumor sphericity across WHO histologic grades (Grade II vs. Grade III). (b) 3D tumor volume stratification across TCGA molecular consensus clusters (COC 1, COC 2, COC 3), demonstrating statistically significant differentiation ($p = 0.0268$) with consistent biological rank ordering preservation.

variables were utilized strictly for post-hoc validation and were never seen during model training, feature extraction, or hyperparameter selection:

6.3.1. WHO Histologic Grade Differentiation (Grade II vs. III)

In clinical neuro-oncology, diffuse LGG grade is governed by microscopic cellular atypia and mitotic index rather than macroscopic tumor burden. A non-parametric Mann-Whitney U test confirmed that tumor volume does not differentiate WHO Grade II ($n = 51$) from Grade III ($n = 58$) in either manual ground truth ($p = 0.8104$; Grade II mean: 5,928.3 voxels; Grade III mean: 6,120.2 voxels) or CAL-3DH segmentations ($p = 0.8151$; Grade II mean: 6,241.8 voxels; Grade III mean: 6,125.9 voxels). In contrast, 3D sphericity Ψ revealed a statistically significant distinction in radiologist ground truth (Grade II: 0.5316 ± 0.1087, median 0.5528 [IQR: 0.4381–0.6036]; Grade III: 0.4718 ± 0.0930, median 0.4904 [IQR: 0.4067–0.5309], Mann-Whitney $U = 1989.0, p = 0.0020$), reflecting greater boundary irregularity in higher-grade neoplasms. For CAL-3DH segmentations, sphericity preserved this aligned directional reduction (Grade II: 0.4910 ± 0.1184, median 0.4776 [IQR: 0.4107–0.5753]; Grade III: 0.4623 ± 0.1106, median 0.4711 [IQR: 0.3905–0.5450], Mann-Whitney $U = 1637.0, p = 0.3388$). While CAL-3DH faithfully preserves the directional reduction in 3D sphericity from Grade II to Grade III, statistical significance is attenuated in automated masks ($p = 0.3388$ vs. $p = 0.0020$ in ground truth) due to boundary discretization and surface staircasing on the standardized 128×128 voxel grid.

6.3.2. Molecular Subtype Stratification (COCCluster)

The TCGA Pan-Glioma consortium stratifies LGG into three distinct Cluster-of-Clusters (COC) molecular platforms: COC 1 ($n = 56$, IDH-wildtype-like tumors exhibiting clinically

aggressive growth), COC 2 ($n = 24$, IDH-mutant non-codel astrocytomas), and COC 3 ($n = 30$, IDH-mutant 1p/19q co-deleted oligodendrogliomas).

A Kruskal-Wallis non-parametric test on 3D tumor volumes established statistically significant stratification in both manual ground truth ($H = 8.16, p = \mathbf{0.0169}$, effect size $\epsilon^2 = 0.0576$) and automated CAL-3DH segmentations ($H = 7.24, p = \mathbf{0.0268}$, effect size $\epsilon^2 = 0.0489$). Group comparisons confirm that the reported volume values represent cohort group means preserving consistent biological rank ordering:

$$\text{COC 1 Mean (7,098.0)} > \text{COC 3 Mean (5,509.6)} > \text{COC 2 Mean (4,834.5)} \tag{24}$$

Specifically, ground truth volumes yielded: COC 1 mean $6{,}780.5 \pm 4{,}816.2$ voxels (median 6,278.0, IQR: 2,546.5–8,825.5); COC 2 mean $4{,}390.2 \pm 5{,}226.1$ voxels (median 2,680.0, IQR: 1,243.5–4,709.2); and COC 3 mean $5{,}930.7 \pm 4{,}656.1$ voxels (median 3,857.5, IQR: 1,983.0–8,776.0). For CAL-3DH, the segmented volumes yielded: COC 1 mean $7{,}098.0 \pm 5{,}082.1$ voxels (median 6,535.5, IQR: 2,748.5–9,582.5); COC 2 mean $4{,}834.5 \pm 5{,}690.1$ voxels (median 2,566.5, IQR: 1,492.8–5,467.0); and COC 3 mean $5{,}509.6 \pm 4{,}126.6$ voxels (median 4,537.0, IQR: 2,507.2–7,989.0). Pairwise post-hoc Mann-Whitney U testing confirmed that volumetric separation between COC 1 and COC 2 ($U = 926.0$, unadjusted $p = 0.0078$) remains statistically significant after Bonferroni correction across the three pairwise cluster comparisons ($p_{\text{Bonf}} = 3 \times 0.0078 = 0.0234 < 0.05$), with expected moderate overlap between intermediate clusters (COC 1 vs. COC 3, $p = 0.2128, p_{\text{Bonf}} = 0.6384$; COC 3 vs. COC 2, $p = 0.2007, p_{\text{Bonf}} = 0.6021$).

7. Discussion and Benchmark Comparison

7.1. Comparison with State-of-the-Art Literature

Table 8 benchmarks our results against published investigations evaluated strictly on the TCGA-LGG benchmark dataset (Buda et al., 2019).

In Table 8, we benchmark CAL-3DH alongside published studies on the TCGA-LGG cohort. Notably, Buda et al. (2019) reported a patient-level volumetric Dice mean of 82% (median 85%) using a standard 2D U-Net evaluated at 256×256 resolution under their patient-level split. Under our strict 5-fold nested (75/13/22) zero-leakage protocol at standardized 128×128 resolution, CAL-3DH achieves a comparable median 3D volumetric Dice of 84.42% (mean 77.73%, $HD_{95} = 10.97$ voxels, $\text{ICC}(2,1) = 0.8894$) alongside probability calibration and empty-slice false-alarm suppression. Furthermore, recent investigations on the TCGA-LGG benchmark such as LiteMRINet (Yuan and Kang, 2025) (85.50% soft Dice) and PoSAM-ULTRA (Alhatamleh et al., 2026) (91.40% soft Dice) evaluate under random slice-level partitioning. As demonstrated by our controlled audit (Section 5.5), random slice partitioning induces substantial performance inflation (+8.79 pp on positive-slice Dice and +6.20 pp on overall Dice) because contiguous slices from the same brain volume leak into both training and evaluation subsets, allowing networks to memorize subject anatomy. When audited under identical random-slice conditions, our baseline architecture achieves an overall Dice of 88.22% (global soft Dice of 88.18%) and reaches 89.83% under threshold

Table 8: Systematic benchmark comparison against published deep learning studies on the TCGA-LGG dataset, stratified by data partitioning protocol and evaluation metric scope.

Study	Year	Architecture	Post-Proc. / Calibration	Params	Overall Dice_{all} (%)	Tumor Dice_{pos} (%)	3D Vol. Dice / ICC
Part A: Random-Slice Partitioning (Data-Leaked Regime)							
LiteMRINet (Yuan and Kang, 2025)	2025	LiteMRINet	None / None	1.61M	85.50 (Soft)†	NR	NR
PoSAM-ULTRA (Alhatamleh et al., 2026)	2026	PoSAM-ULTRA	Prompt Tuning	NR	91.40 (Soft)†	NR	NR
This Work (Fixed $\tau = 0.50$)	2026	Attn Res-UNet	None / None	8.20M	88.22 [Soft: 88.18]†	82.74†	Invalid (Leaked)‡
This Work (Tuned τ^*)	2026	Attn Res-UNet	None / Inner τ^*	8.20M	89.83†	83.75†	Invalid (Leaked)‡
Part B: Patient-Grouped Partitioning (Zero-Leakage Clinical Standard)							
Buda et al. (Buda et al., 2019)	2019	Standard 2D U-Net	None / Fixed $\tau = 0.50$	7.80M	NR	NR	82.00% (Med: 85.0%)[a] / NR
This Work (2D Baseline)	2026	Attn Res-UNet	None / Fixed $\tau = 0.50$	8.20M	82.02	73.96	76.00% / 0.8652
This Work (2D Nested)	2026	Attn Res-UNet	Temp (T^*) / Nested τ_k^*	8.20M	84.45	74.60	77.57% / 0.8860
CAL-3DH (Ours)	**2026**	**Attn Res-UNet**	**Temp (T^*) / 3D Hyst.**	**8.20M**	**86.12**	**74.92**	**77.73% / 0.8894**

[a] Buda et al. (2019) reported a patient-level volumetric Dice mean of 82% (median 85%) evaluated at 256×256 resolution under their specific split; under our strict 5-fold nested (75/13/22) zero-leakage protocol at standardized 128×128 resolution, CAL-3DH achieves a comparable median 3D volumetric Dice of 84.42% (mean 77.73%) alongside probability calibration and empty-slice false-alarm suppression.

†Performance inflated by slice-level data leakage (inter-slice correlation and subject feature memorization across training and testing partitions).

‡Patient-level 3D evaluation is methodologically invalid in random-slice partitioning because slices from each patient are split across training and test sets.

NR: Not Reported.

calibration, directly confirming that elevated scores reflect partition design rather than architectural superiority. Moreover, differences in data partitioning protocols and empty-slice Dice inclusion conventions preclude direct numerical equivalence. Rather than claiming unqualified algorithmic superiority over leaked models, Table 8 primarily serves to illuminate the substantial methodological divergence between random-slice and zero-leakage patient-grouped regimes.

7.2. Limitations and Future Scope

Five primary limitations warrant consideration. First, while the TCGA-LGG cohort represents a multi-institutional benchmark, prospective validation across external centers with heterogeneous scanner field strengths (1.5 T vs. 3.0 T) and varying acquisition protocols will further substantiate generalizability. Future studies should also implement center-stratified cross-validation using institution codes where DICOM headers permit. Second, the public TCGA-LGG TIFF benchmark distribution lacks raw DICOM/NIfTI physical millimeter spacing headers, necessitating evaluation in normalized voxel space; acquiring full volumetric DICOM metadata in future cohorts will enable isotropic millimeter re-sampling. Third, our cross-validation utilized a fixed single seed (42); while extensive 10,000-sample bootstrap confidence intervals verify metric stability, multi-seed repetitions across massive multi-center registries will provide deeper insight into initialization variance, and full end-to-end retraining ablations isolating individual preprocessing stages (such as wavelet denoising vs. raw input training) represent an additional direction for future work. Fourth, in accordance with Sex and Gender Equity in Research (SAGER) and SGBA reporting guidelines, while demographic sex distributions were documented across the cohort (50.9% male, 48.2% female), sex- and gender-stratified subgroup calibration was not performed due to sample size constraints ($N = 110$ patients); evaluating potential sex-dependent morphological calibration shifts represents an important direction for larger cohorts. Fifth, while CAL-3DH preserved the directional reduction in 3D sphericity from Grade II (0.4910) to Grade III (0.4623), statistical significance was attenuated in automated masks ($p = 0.3388$ vs. $p = 0.0020$ in ground truth) due to boundary discretization and minor surface staircasing on the standardized 128×128 voxel grid; future investigations with sub-voxel contour refinement and continuous mesh-based surface representations will be valuable for capturing subtle morphological biomarkers.

8. Conclusion

The clinical translation of automated deep learning systems for lower-grade glioma management requires rigorous adherence to methodological integrity, reliable uncertainty quantification, and anatomical continuity across volumetric scans. In this study, we addressed the pervasive vulnerabilities of 2D segmentation pipelines through CAL-3DH, an end-to-end framework uniting leakage auditing, nested calibration, and anisotropic volumetric post-processing.

Our controlled audit empirically demonstrated that conventional slice-level random splitting induces a substantial +8.79 pp (+11.88% relative) artificial Dice inflation by allowing

deep networks to memorize patient-specific morphological features across contiguous slices rather than extracting generalizable neoplastic patterns. Furthermore, we showed that uncalibrated overlap losses generate overconfident probabilistic halos that can be effectively regularized via nested temperature scaling without shifting decision boundaries. By explicitly decoupling in-plane planar binarization from axial boundary rescue along the z-axis, the CAL-3DH algorithm eliminated 42.4% of false alarms on tumor-free tissue while recovering infiltrative apical margins. Beyond geometric metrics, the extracted 3D volumes demonstrated good-to-excellent reliability against radiologist assessments ($\mathrm{ICC}(2,1) = 0.8894$, 95% CI: $[0.7782, 0.9521]$) and preserved statistically significant stratification across TCGA-defined molecular consensus clusters ($p = 0.0268$).

By establishing a transparent, leakage-audited, patient-grouped benchmark on the complete TCGA-LGG cohort, CAL-3DH provides a standardized methodological template and a robust volumetric framework to support surgical planning, radiation oncology, and quantitative neuro-oncology diagnostics.

CRediT authorship contribution statement

Shayan Ganji Nataj: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Keivan Borna:** Supervision, Project administration, Writing – review and editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data statement

The TCGA-LGG MRI collection is openly accessible via The Cancer Imaging Archive (TCIA) (Clark et al., 2013; Pedano et al., 2016) and mirrored on Kaggle at https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation. The complete CAL-3DH open-source codebase, patient-grouped 5-fold cross-validation splits, precomputed caching modules, model weights, and leakage auditing scripts are openly hosted on GitHub at https://github.com/Shayan-Ganji/lgg_segmentation.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) solely for language editing, grammatical polishing, and code debugging assistance. All experimental designs, mathematical formulations, statistical audits, and scientific interpretations were conducted by the authors. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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